

```

%root_path = fileparts(mfilename("fullpath"));
%root_path = pwd;

% retrieve selected cuts used for further fitting
GP_FFCutsFits =
recover_selected_cuts('GP_FFCutsFits','e:\SHARE\FE\Data\analysis\07_fit_with_multicu
t_allJ\sel_cuts');

```

```

fn = fieldnames(GP_FFCutsFits);
%nf = 11
%for nf = 2:numel(fn)
%selectCuts = fn{nf};
selectCuts = 'Ei800_off200GPminFF';

```

```

selectCuts =
'Ei800_off200GPminFF'

```

```

if ismember(selectCuts(1:3),{'msk','dat'})||...
    contains(selectCuts,'all_fit_par')
    %continue
    error('Can not fit field: %s',selectCuts);
end

```

```

fprintf('Processing cuts: %s \n', selectCuts);

```

```

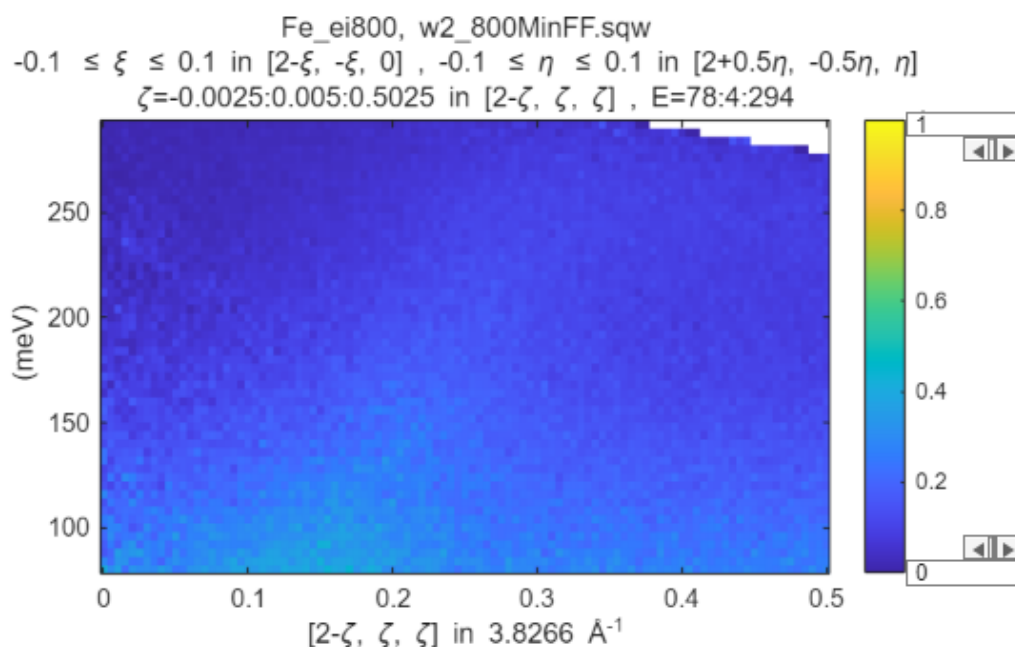
Processing cuts: Ei800_off200GPminFF

```

```

%selectCuts= 'Ei400_off200GPmaxFF';
cuts_list = GP_FFCutsFits.(selectCuts);
if ~iscell(cuts_list)
    cuts_list = {cuts_list};
end
if contains(selectCuts,'Ei200')
    mi = maps_instrument(200,600,'S');
end
if contains(selectCuts,'Ei400')
    mi = maps_instrument(401,600,'S');
end
if contains(selectCuts,'Ei800')
    mi = maps_instrument(786,600,'S');
end
sample=IX_sample(true,[1,0,0],[0,1,0],'cuboid',[0.04,0.03,0.02]);
for j=1:numel(cuts_list)
    cuts_list{j} = cuts_list{j}.set_instrument(mi);
    cuts_list{j} = cuts_list{j}.set_sample(sample);
    plot(cuts_list{j}); lz 0 1; keep_figure;
end

```



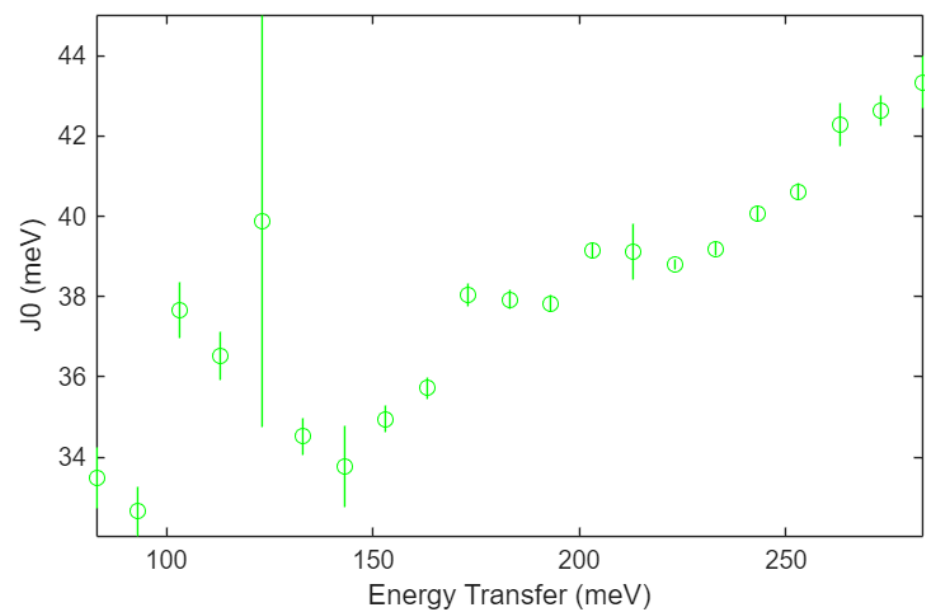
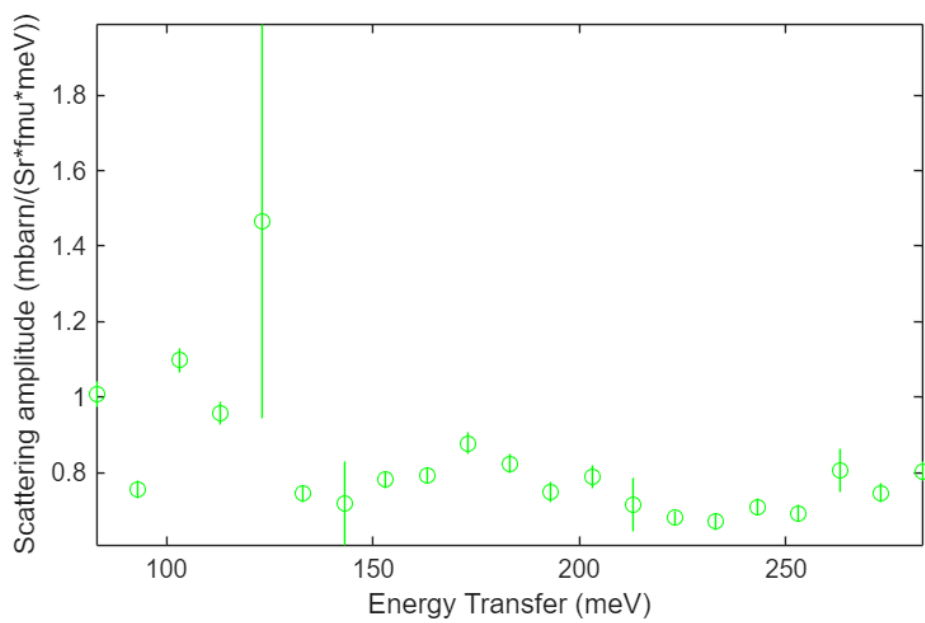
```
dir_name = "GP";

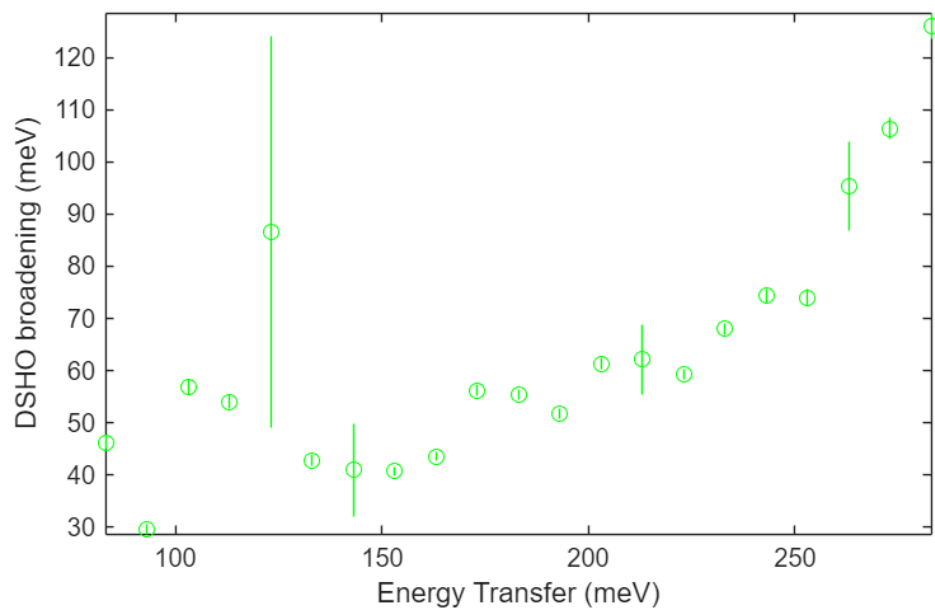
dE_step = 2; %original energy transfer step data were binned to. No point in going
finer
half_dE = 5; % half width of data binning

en_mm = cuts_list{1}.data.img_range(:,4);
en_mm(1) = max(30,en_mm(1));
%en_mm(1) = 100;
%en_mm(2) = 170;
cut_en = en_mm(1)+half_dE:2*half_dE:(en_mm(2)-half_dE);

cut_dimensions = 1;
GP_FFCutsFits.(selectCuts) = cuts_list;

GP_FFCutsFits = fit_multicutsFF_along_direction(...
    GP_FFCutsFits,selectCuts,cut_dimensions,cut_en,dE_step,half_dE,true);
```

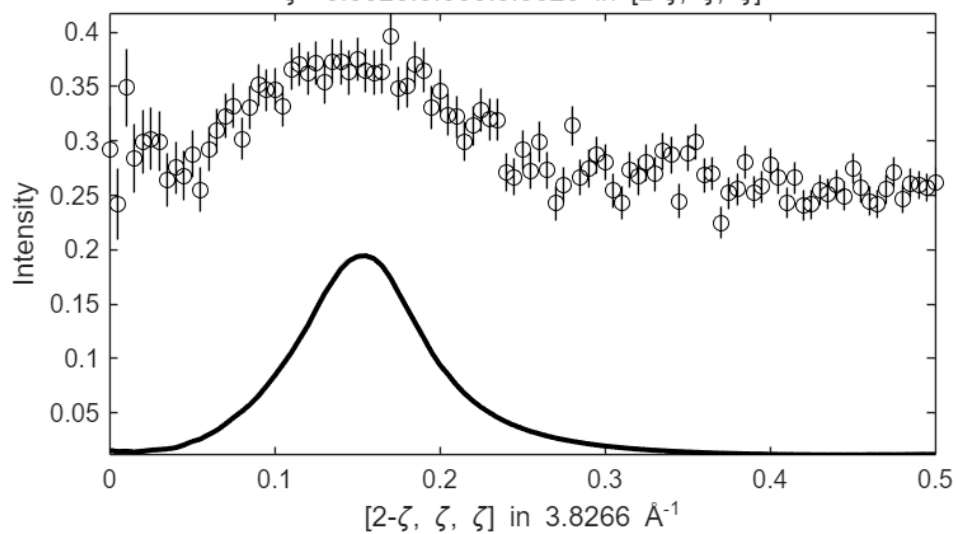




 **** En = 83±5 going to 283

Fe_ei800, w2_800MinFF.sqw

$0.1 \leq \xi \leq 0.1$ in $[2-\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2+0.5\eta, -0.5\eta, \eta]$, $78 \leq E \leq$
 $\zeta=-0.0025:0.005:0.5025$ in $[2-\zeta, \zeta, \zeta]$



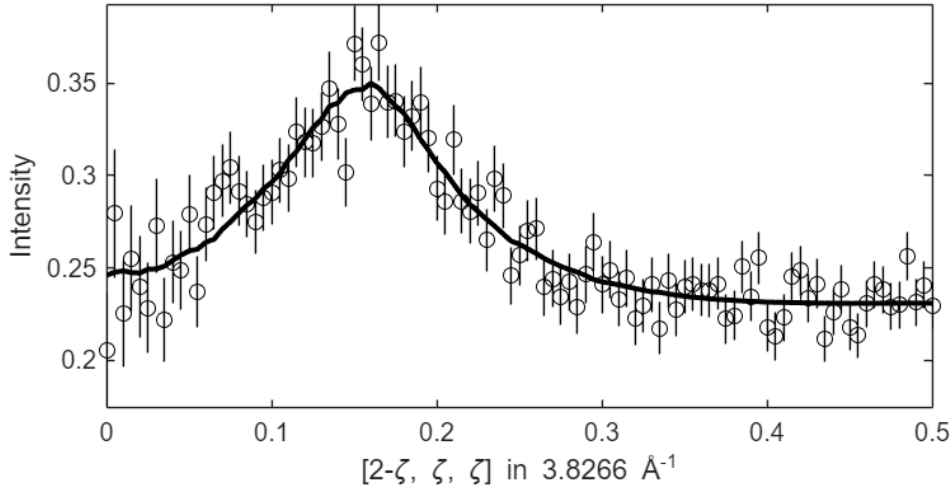
 **** En = 93±5 going to 283

Fe_ei800, w2_800MinFF.sqw

S=0.74±0.022; J0= 33±0.61; J1= 0± 0; gamma=28.6685±0.8532

0.1 ≤ ξ ≤ 0.1 in [2-ξ, -ξ, 0] , -0.1 ≤ η ≤ 0.1 in [2+0.5η, -0.5η, η] , 88 ≤ E ≤

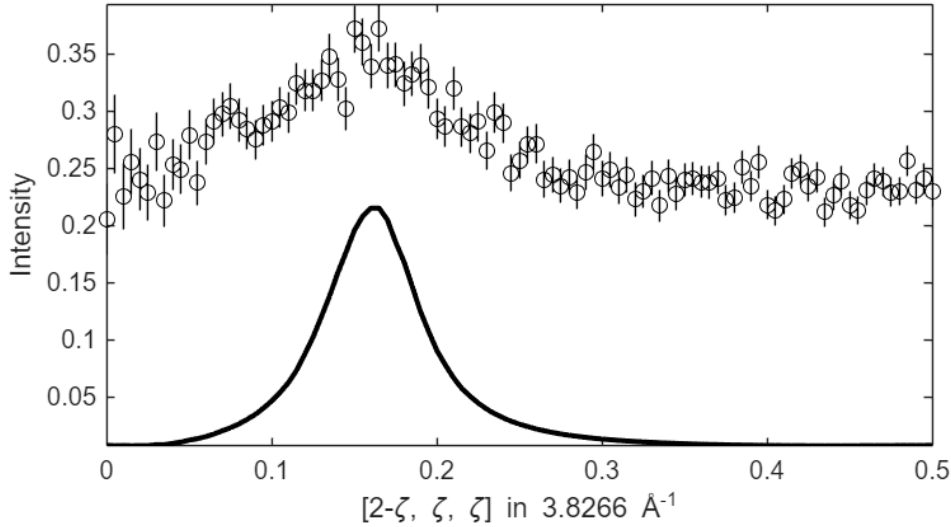
ζ=-0.0025:0.005:0.5025 in [2-ζ, ζ, ζ]



Fe_ei800, w2_800MinFF.sqw

0.1 ≤ ξ ≤ 0.1 in [2-ξ, -ξ, 0] , -0.1 ≤ η ≤ 0.1 in [2+0.5η, -0.5η, η] , 88 ≤ E ≤

ζ=-0.0025:0.005:0.5025 in [2-ζ, ζ, ζ]



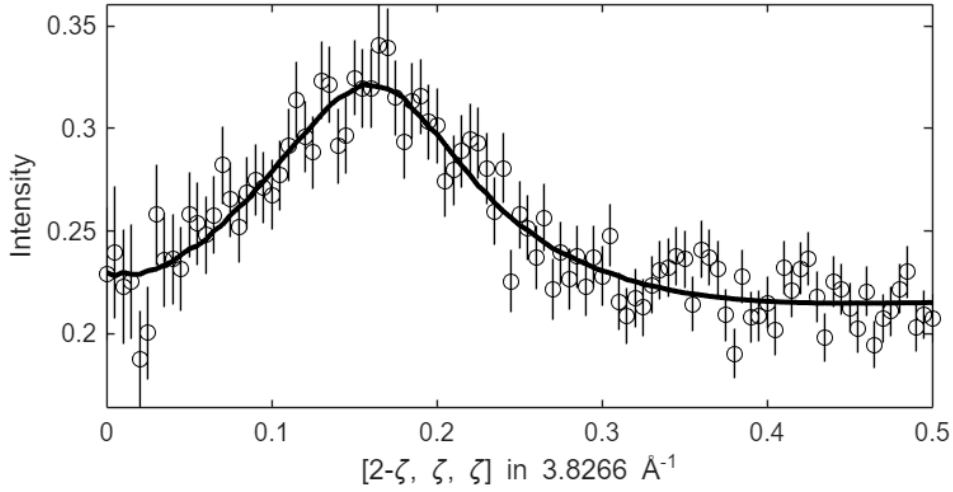
**** En = 103±5 going to 283

Fe_ei800, w2_800MinFF.sqw

S= 1.1±0.032; J0= 37± 0.7; J1= 0± 0; gamma=54.1741±1.33717

0.1 ≤ ξ ≤ 0.1 in [2-ξ, -ξ, 0] , -0.1 ≤ η ≤ 0.1 in [2+0.5η, -0.5η, η] , 98 ≤ E ≤

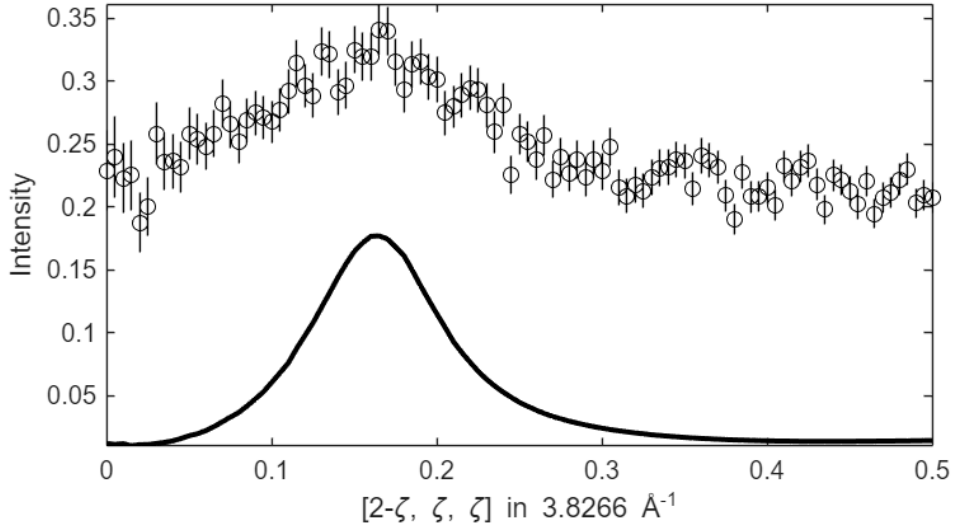
ζ=-0.0025:0.005:0.5025 in [2-ζ, ζ, ζ]



Fe_ei800, w2_800MinFF.sqw

0.1 ≤ ξ ≤ 0.1 in [2-ξ, -ξ, 0] , -0.1 ≤ η ≤ 0.1 in [2+0.5η, -0.5η, η] , 98 ≤ E ≤

ζ=-0.0025:0.005:0.5025 in [2-ζ, ζ, ζ]



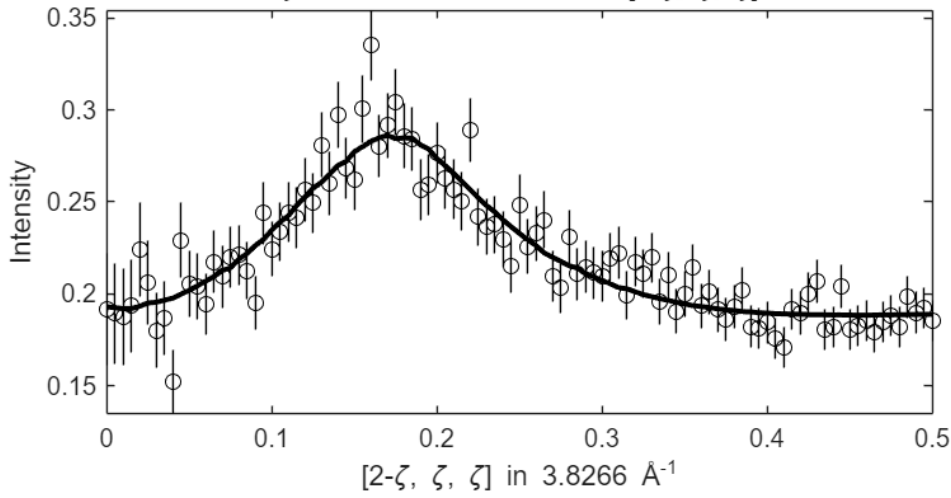
**** En = 113±5 going to 283

Fe_ei800, w2_800MinFF.sqw

S=0.97±0.03; J0= 37±0.62; J1= 0± 0; gamma=54.8823±1.35879

.1 ≤ ξ ≤ 0.1 in [2-ξ, -ξ, 0] , -0.1 ≤ η ≤ 0.1 in [2+0.5η, -0.5η, η] , 108 ≤ E ≤

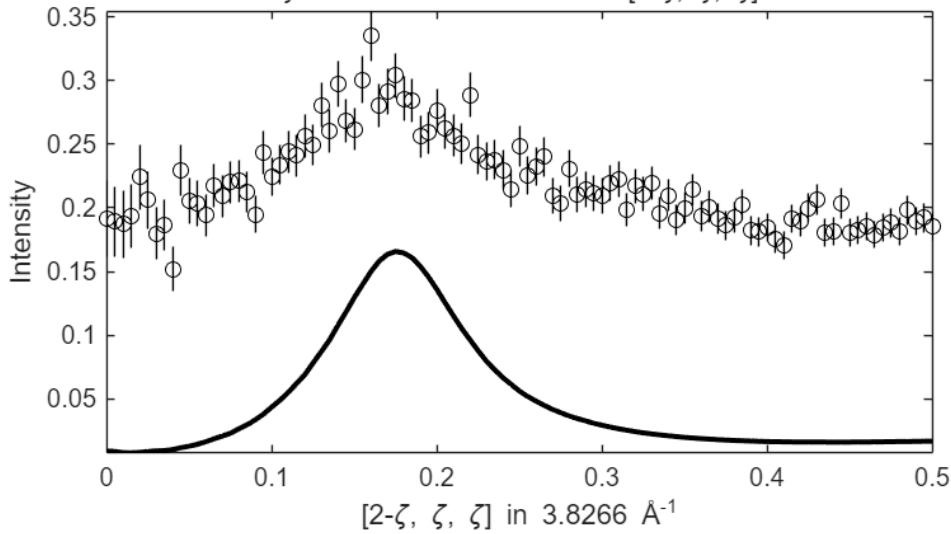
ζ=-0.0025:0.005:0.5025 in [2-ζ, ζ, ζ]



Fe_ei800, w2_800MinFF.sqw

.1 ≤ ξ ≤ 0.1 in [2-ξ, -ξ, 0] , -0.1 ≤ η ≤ 0.1 in [2+0.5η, -0.5η, η] , 108 ≤ E ≤

ζ=-0.0025:0.005:0.5025 in [2-ζ, ζ, ζ]



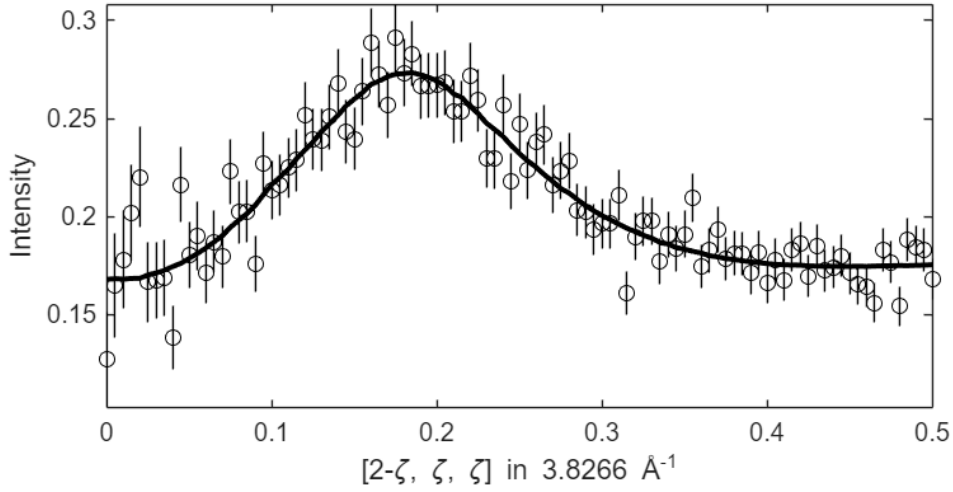
**** En = 123±5 going to 283

Fe_ei800, w2_800MinFF.sqw

S= 1.5±0.049; J0= 40±0.54; J1= 0± 0; gamma=86.304±1.64742

.1 ≤ ξ ≤ 0.1 in [2-ξ, -ξ, 0] , -0.1 ≤ η ≤ 0.1 in [2+0.5η, -0.5η, η] , 118 ≤ E ≤

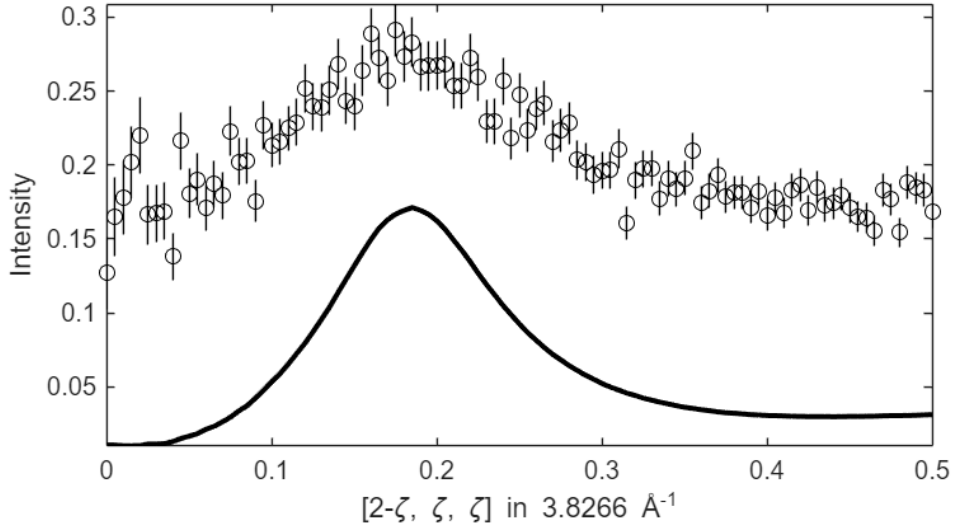
ζ=-0.0025:0.005:0.5025 in [2-ζ, ζ, ζ]



Fe_ei800, w2_800MinFF.sqw

.1 ≤ ξ ≤ 0.1 in [2-ξ, -ξ, 0] , -0.1 ≤ η ≤ 0.1 in [2+0.5η, -0.5η, η] , 118 ≤ E ≤

ζ=-0.0025:0.005:0.5025 in [2-ζ, ζ, ζ]



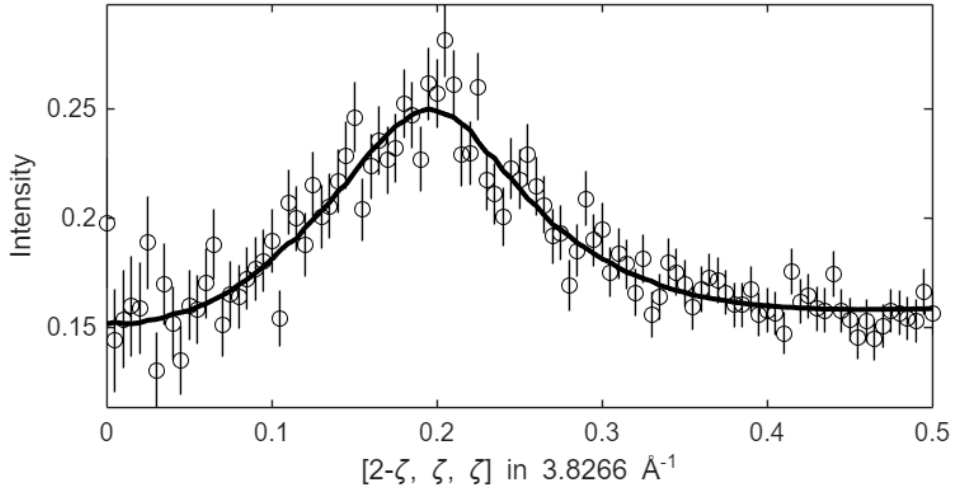
**** En = 133±5 going to 283

Fe_ei800, w2_800MinFF.sqw

S=0.75±0.14; J0= 35± 1.4; J1= 0± 0; gamma=43.1561±11.4598

.1 ≤ ξ ≤ 0.1 in [2-ξ, -ξ, 0] , -0.1 ≤ η ≤ 0.1 in [2+0.5η, -0.5η, η] , 128 ≤ E ≤

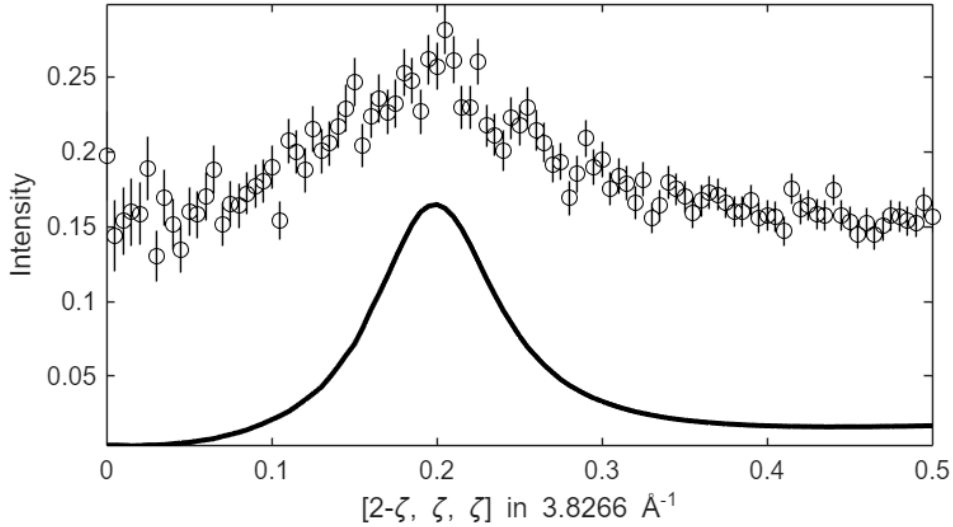
ζ=-0.0025:0.005:0.5025 in [2-ζ, ζ, ζ]



Fe_ei800, w2_800MinFF.sqw

.1 ≤ ξ ≤ 0.1 in [2-ξ, -ξ, 0] , -0.1 ≤ η ≤ 0.1 in [2+0.5η, -0.5η, η] , 128 ≤ E ≤

ζ=-0.0025:0.005:0.5025 in [2-ζ, ζ, ζ]



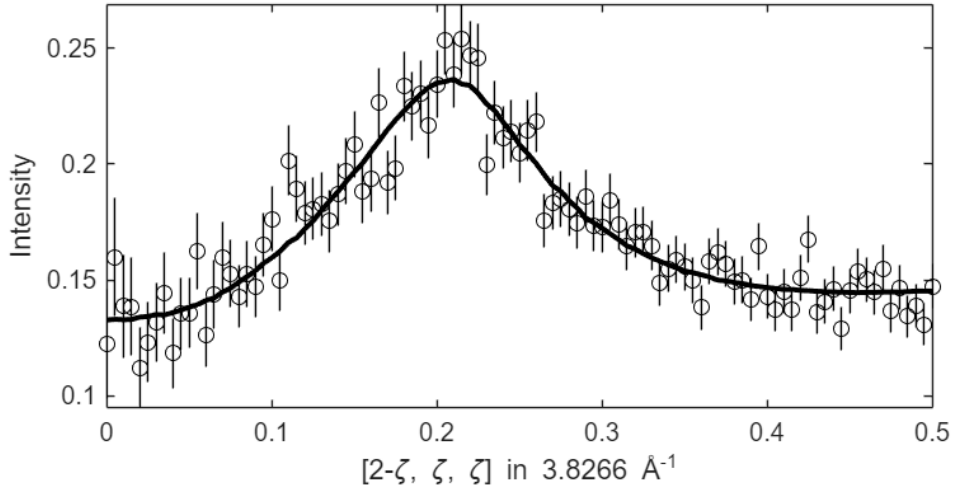
**** En = 143±5 going to 283

Fe_ei800, w2_800MinFF.sqw

S=0.73±0.021; J0= 34±0.37; J1= 0± 0; gamma=42.0677±0.907378

.1 ≤ ξ ≤ 0.1 in [2-ξ, -ξ, 0] , -0.1 ≤ η ≤ 0.1 in [2+0.5η, -0.5η, η] , 138 ≤ E ≤

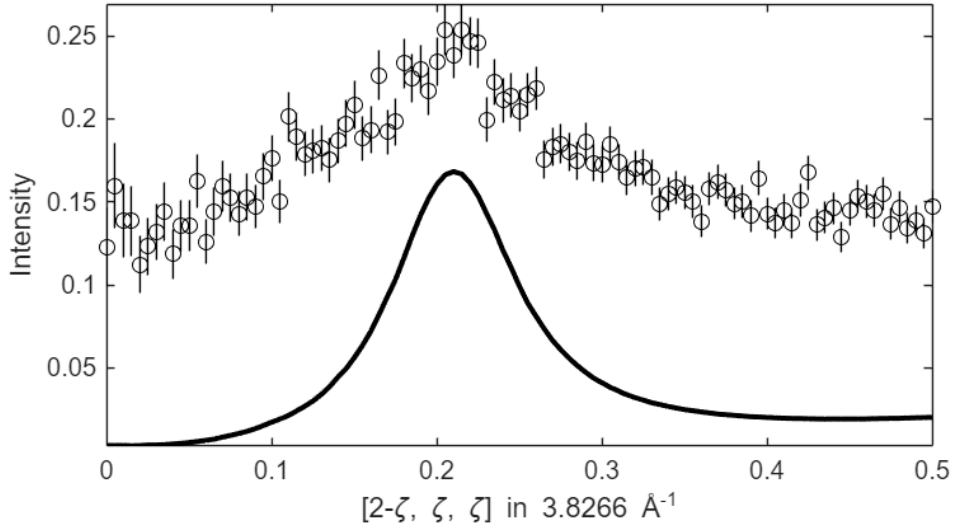
ζ=-0.0025:0.005:0.5025 in [2-ζ, ζ, ζ]



Fe_ei800, w2_800MinFF.sqw

.1 ≤ ξ ≤ 0.1 in [2-ξ, -ξ, 0] , -0.1 ≤ η ≤ 0.1 in [2+0.5η, -0.5η, η] , 138 ≤ E ≤

ζ=-0.0025:0.005:0.5025 in [2-ζ, ζ, ζ]



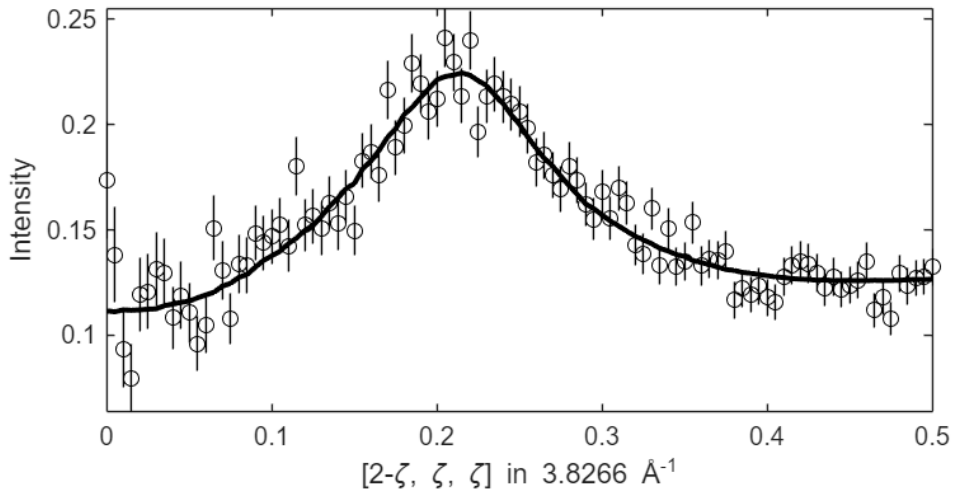
**** En = 153±5 going to 283

Fe_ei800, w2_800MinFF.sqw

S=0.75±0.021; J0= 35±0.34; J1= 0± 0; gamma=39.2145±0.835738

.1 ≤ ξ ≤ 0.1 in [2-ξ, -ξ, 0] , -0.1 ≤ η ≤ 0.1 in [2+0.5η, -0.5η, η] , 148 ≤ E ≤

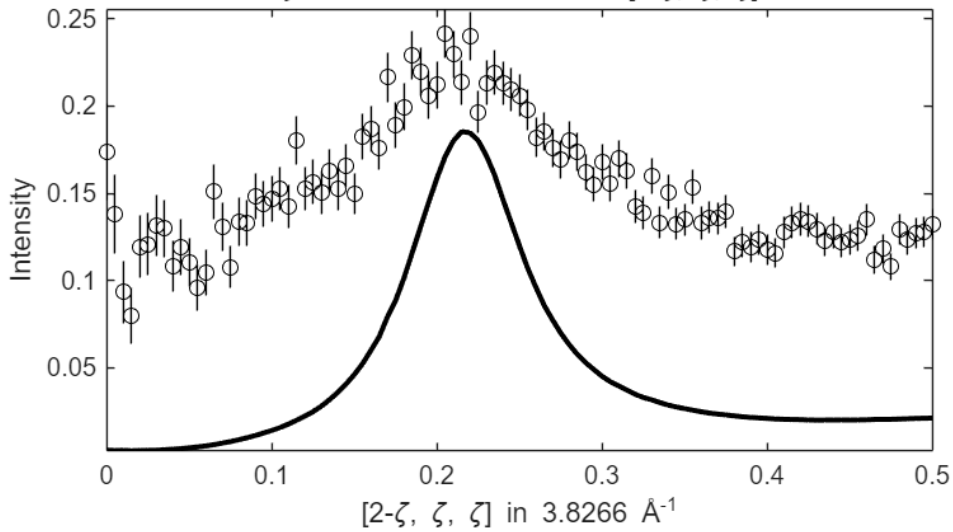
ζ=-0.0025:0.005:0.5025 in [2-ζ, ζ, ζ]



Fe_ei800, w2_800MinFF.sqw

.1 ≤ ξ ≤ 0.1 in [2-ξ, -ξ, 0] , -0.1 ≤ η ≤ 0.1 in [2+0.5η, -0.5η, η] , 148 ≤ E ≤

ζ=-0.0025:0.005:0.5025 in [2-ζ, ζ, ζ]

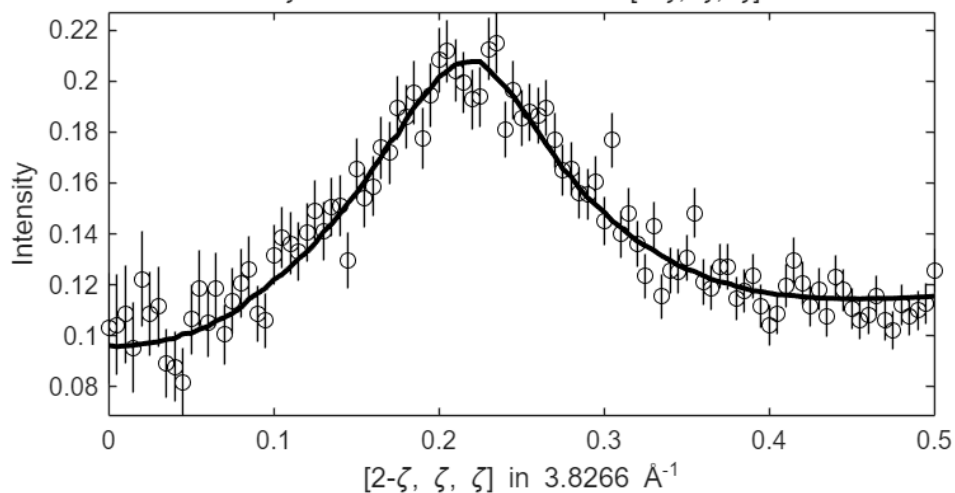


**** En = 163±5 going to 283

Fe_ei800, w2_800MinFF.sqw

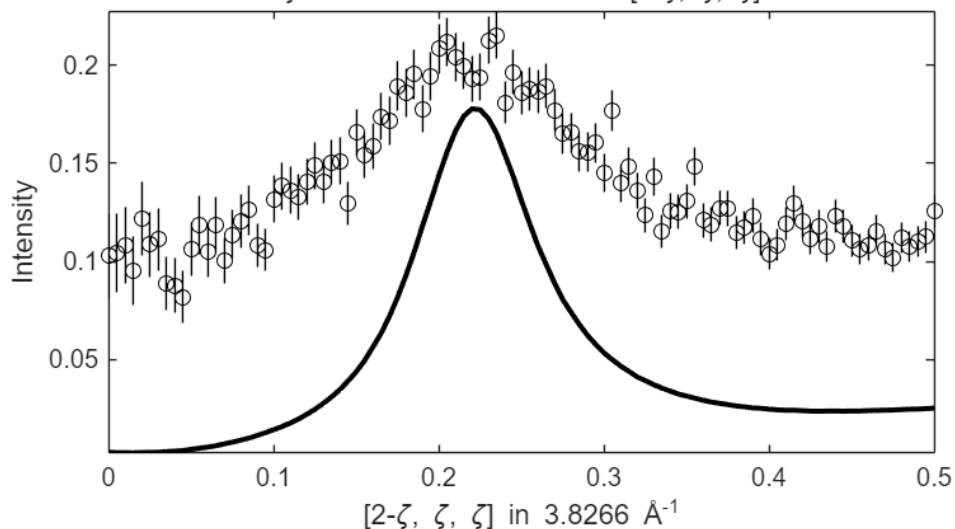
S= 0.8±0.021; J0= 36±0.27; J1= 0± 0; gamma=43.9873±0.778375

.1 ≤ ξ ≤ 0.1 in [2-ξ, -ξ, 0] , -0.1 ≤ η ≤ 0.1 in [2+0.5η, -0.5η, η] , 158 ≤ E ≤
ζ=-0.0025:0.005:0.5025 in [2-ζ, ζ, ζ]



Fe_ei800, w2_800MinFF.sqw

.1 ≤ ξ ≤ 0.1 in [2-ξ, -ξ, 0] , -0.1 ≤ η ≤ 0.1 in [2+0.5η, -0.5η, η] , 158 ≤ E ≤
ζ=-0.0025:0.005:0.5025 in [2-ζ, ζ, ζ]



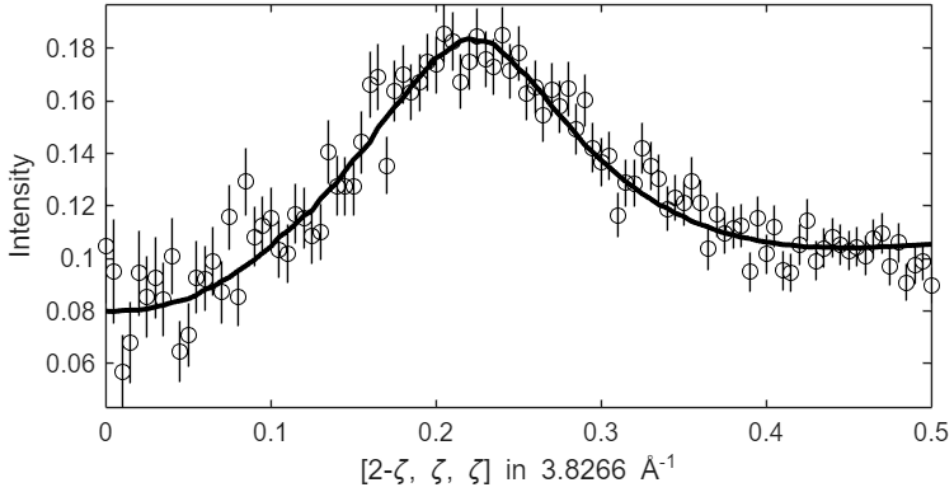
**** En = 173±5 going to 283

Fe_ei800, w2_800MinFF.sqw

S=0.88±0.12; J0= 38±0.98; J1= 0± 0; gamma=56.6943±8.72014

.1 ≤ ξ ≤ 0.1 in [2-ξ, -ξ, 0] , -0.1 ≤ η ≤ 0.1 in [2+0.5η, -0.5η, η] , 168 ≤ E ≤

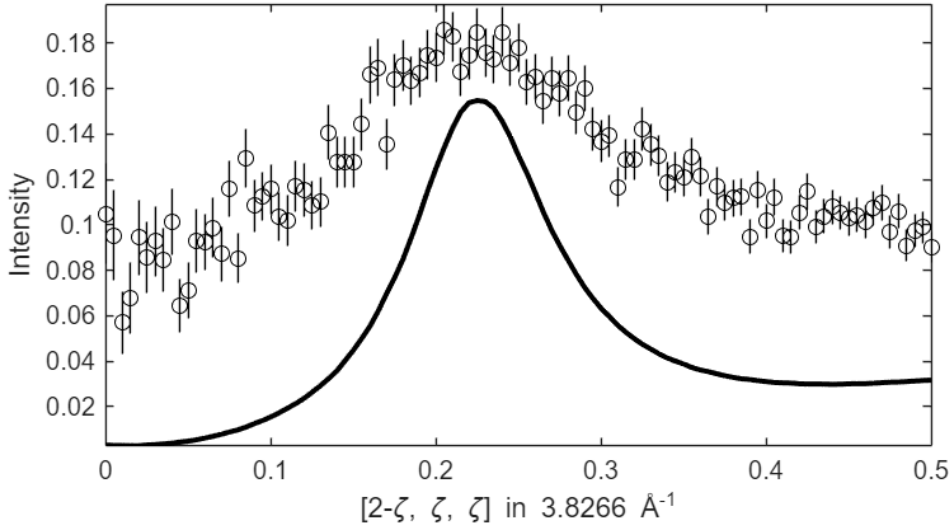
ζ=-0.0025:0.005:0.5025 in [2-ζ, ζ, ζ]



Fe_ei800, w2_800MinFF.sqw

.1 ≤ ξ ≤ 0.1 in [2-ξ, -ξ, 0] , -0.1 ≤ η ≤ 0.1 in [2+0.5η, -0.5η, η] , 168 ≤ E ≤

ζ=-0.0025:0.005:0.5025 in [2-ζ, ζ, ζ]



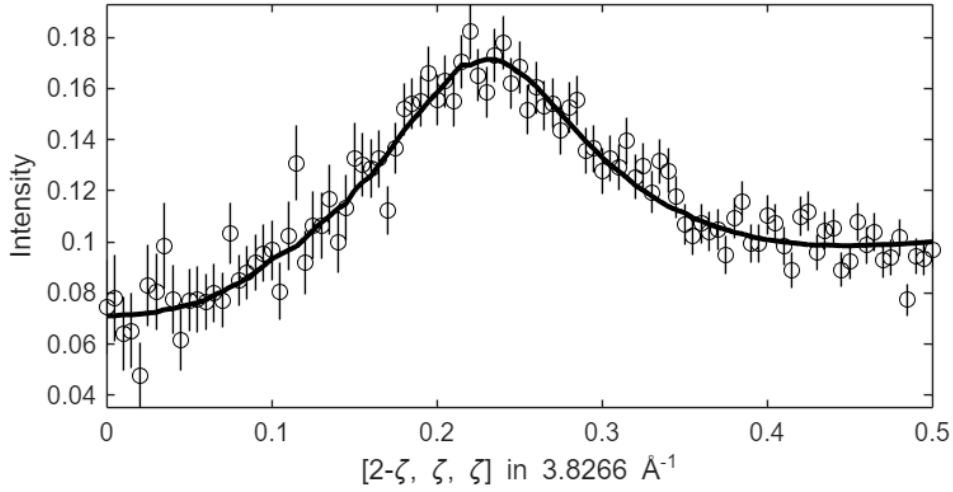
**** En = 183±5 going to 283

Fe_ei800, w2_800MinFF.sqw

S=0.82±0.094; J0= 38±0.83; J1= 0± 0; gamma=55.3696±7.33374

.1 ≤ ξ ≤ 0.1 in [2-ξ, -ξ, 0] , -0.1 ≤ η ≤ 0.1 in [2+0.5η, -0.5η, η] , 178 ≤ E ≤

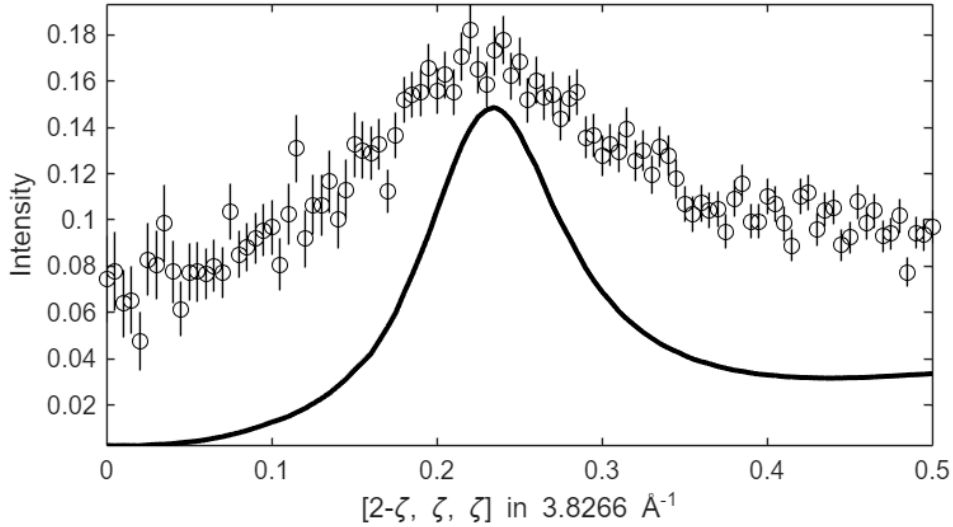
ζ=-0.0025:0.005:0.5025 in [2-ζ, ζ, ζ]



Fe_ei800, w2_800MinFF.sqw

.1 ≤ ξ ≤ 0.1 in [2-ξ, -ξ, 0] , -0.1 ≤ η ≤ 0.1 in [2+0.5η, -0.5η, η] , 178 ≤ E ≤

ζ=-0.0025:0.005:0.5025 in [2-ζ, ζ, ζ]



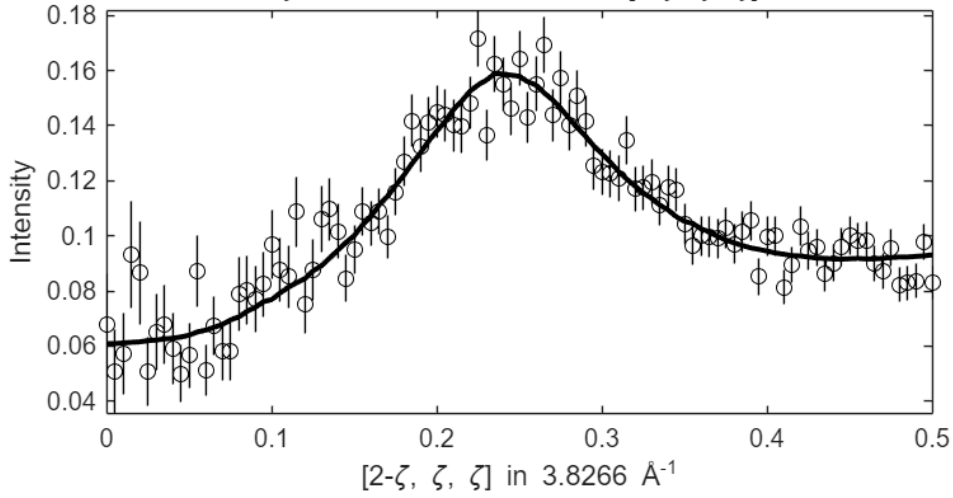
**** En = 193±5 going to 283

Fe_ei800, w2_800MinFF.sqw

S=0.74±0.026; J0= 38±0.21; J1= 0± 0; gamma=51.3373±0.959596

.1 ≤ ξ ≤ 0.1 in [2-ξ, -ξ, 0] , -0.1 ≤ η ≤ 0.1 in [2+0.5η, -0.5η, η] , 188 ≤ E ≤

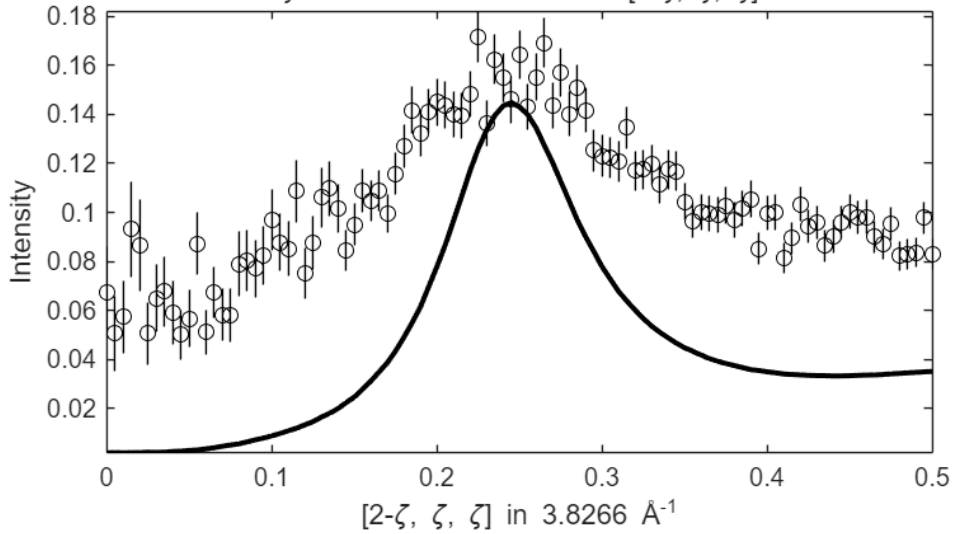
ζ=-0.0025:0.005:0.5025 in [2-ζ, ζ, ζ]



Fe_ei800, w2_800MinFF.sqw

.1 ≤ ξ ≤ 0.1 in [2-ξ, -ξ, 0] , -0.1 ≤ η ≤ 0.1 in [2+0.5η, -0.5η, η] , 188 ≤ E ≤

ζ=-0.0025:0.005:0.5025 in [2-ζ, ζ, ζ]



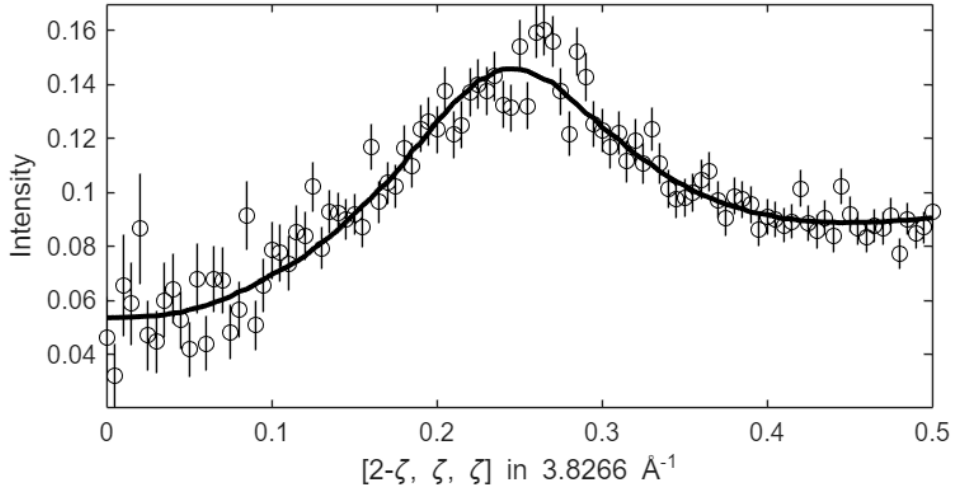
**** En = 203±5 going to 283

Fe_ei800, w2_800MinFF.sqw

S=0.79±0.082; J0= 39±0.75; J1= 0± 0; gamma=61.4656±7.01555

.1 ≤ ξ ≤ 0.1 in [2-ξ, -ξ, 0] , -0.1 ≤ η ≤ 0.1 in [2+0.5η, -0.5η, η] , 198 ≤ E ≤

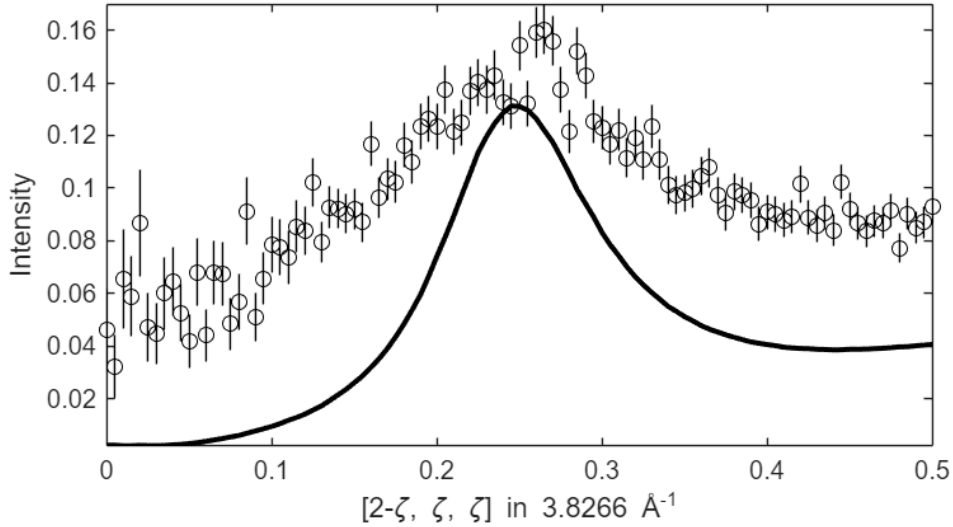
ζ=-0.0025:0.005:0.5025 in [2-ζ, ζ, ζ]



Fe_ei800, w2_800MinFF.sqw

.1 ≤ ξ ≤ 0.1 in [2-ξ, -ξ, 0] , -0.1 ≤ η ≤ 0.1 in [2+0.5η, -0.5η, η] , 198 ≤ E ≤

ζ=-0.0025:0.005:0.5025 in [2-ζ, ζ, ζ]



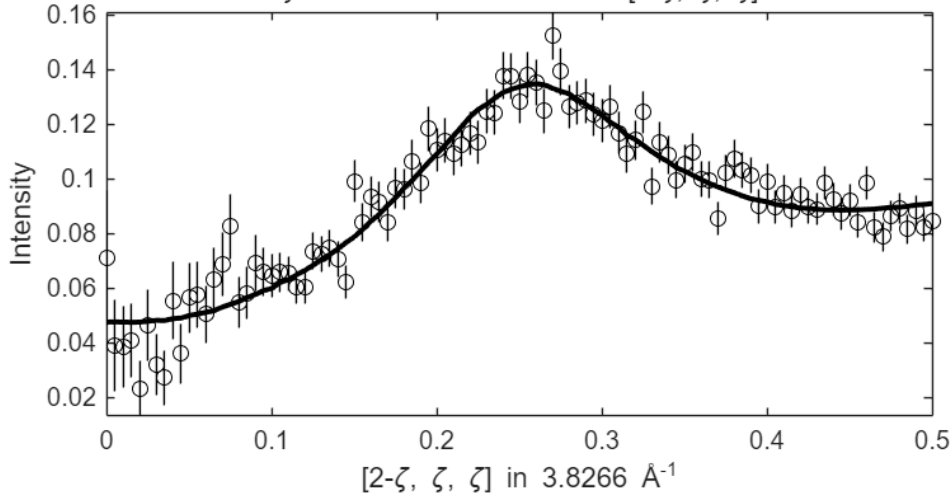
 **** En = 213±5 going to 283

Fe_ei800, w2_800MinFF.sqw

S=0.74±0.073; J0= 39±0.73; J1= 0± 0; gamma=64.5342±7.0366

.1 ≤ ξ ≤ 0.1 in [2-ξ, -ξ, 0] , -0.1 ≤ η ≤ 0.1 in [2+0.5η, -0.5η, η] , 208 ≤ E ≤

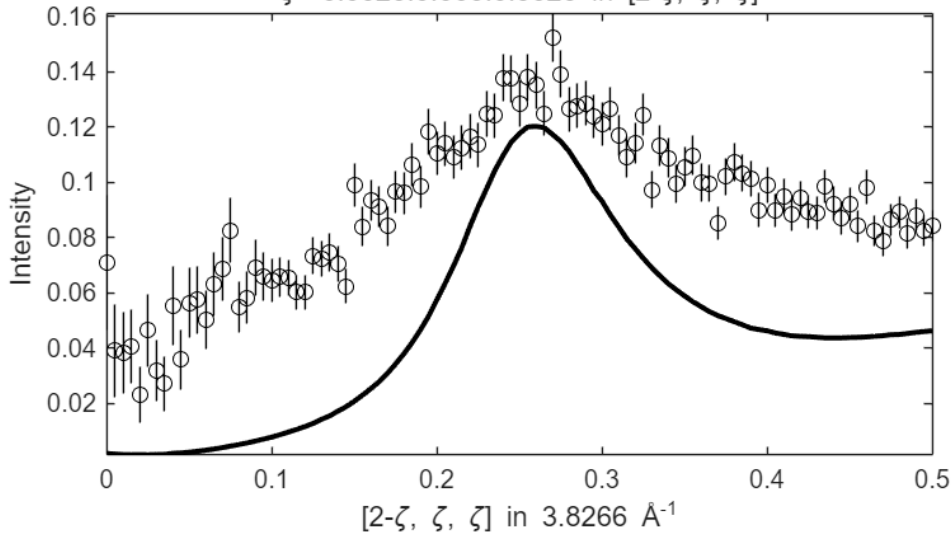
ζ=-0.0025:0.005:0.5025 in [2-ζ, ζ, ζ]



Fe_ei800, w2_800MinFF.sqw

.1 ≤ ξ ≤ 0.1 in [2-ξ, -ξ, 0] , -0.1 ≤ η ≤ 0.1 in [2+0.5η, -0.5η, η] , 208 ≤ E ≤

ζ=-0.0025:0.005:0.5025 in [2-ζ, ζ, ζ]



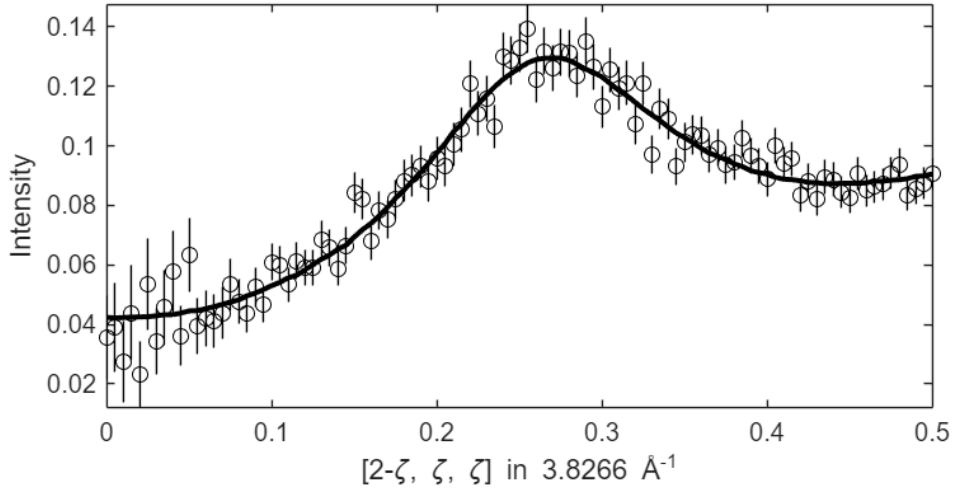
**** En = 223±5 going to 283

Fe_ei800, w2_800MinFF.sqw

S=0.68±0.02; J0= 39±0.13; J1= 0± 0; gamma=59.3672±0.879405

.1 ≤ ξ ≤ 0.1 in [2-ξ, -ξ, 0] , -0.1 ≤ η ≤ 0.1 in [2+0.5η, -0.5η, η] , 218 ≤ E ≤

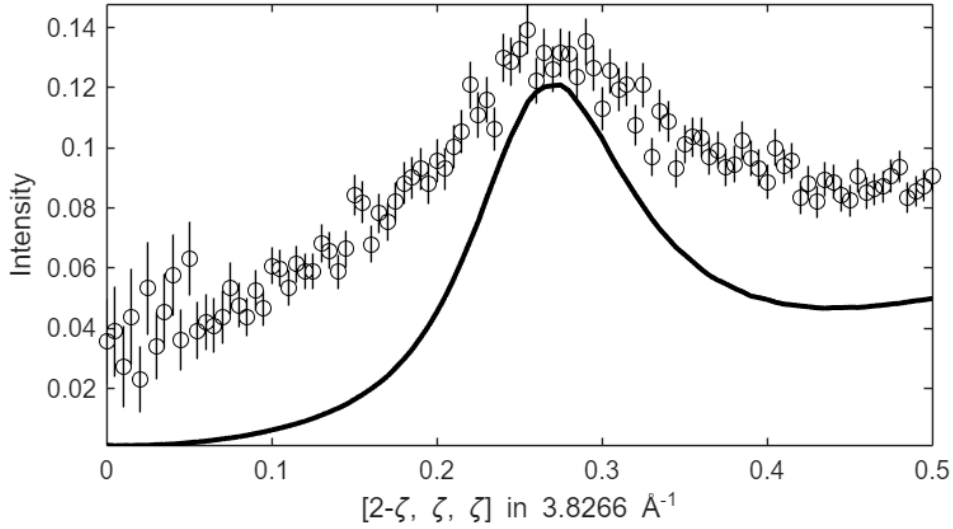
ζ=-0.0025:0.005:0.5025 in [2-ζ, ζ, ζ]



Fe_ei800, w2_800MinFF.sqw

.1 ≤ ξ ≤ 0.1 in [2-ξ, -ξ, 0] , -0.1 ≤ η ≤ 0.1 in [2+0.5η, -0.5η, η] , 218 ≤ E ≤

ζ=-0.0025:0.005:0.5025 in [2-ζ, ζ, ζ]



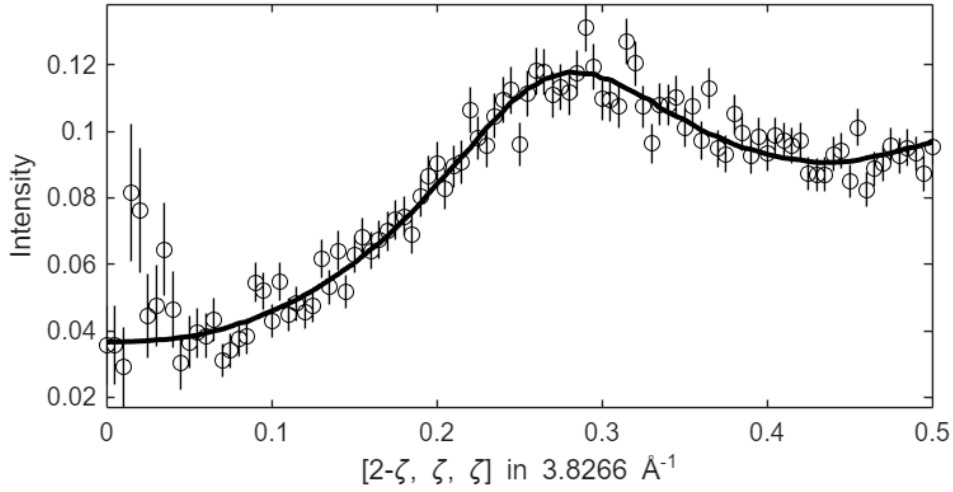
**** En = 233±5 going to 283

Fe_ei800, w2_800MinFF.sqw

S=0.67±0.022; J0= 39±0.16; J1= 0± 0; gamma=67.6729±1.12083

.1 ≤ ξ ≤ 0.1 in [2-ξ, -ξ, 0] , -0.1 ≤ η ≤ 0.1 in [2+0.5η, -0.5η, η] , 228 ≤ E ≤

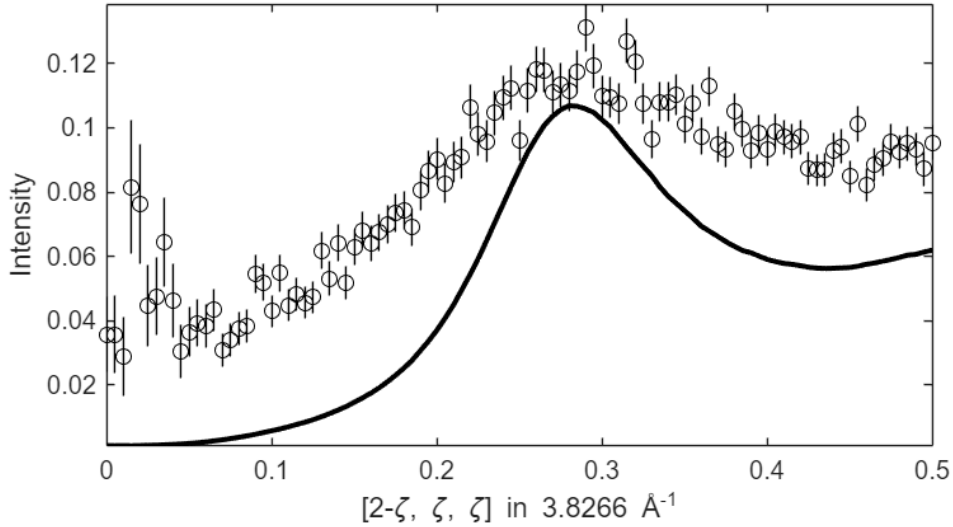
ζ=-0.0025:0.005:0.5025 in [2-ζ, ζ, ζ]



Fe_ei800, w2_800MinFF.sqw

.1 ≤ ξ ≤ 0.1 in [2-ξ, -ξ, 0] , -0.1 ≤ η ≤ 0.1 in [2+0.5η, -0.5η, η] , 228 ≤ E ≤

ζ=-0.0025:0.005:0.5025 in [2-ζ, ζ, ζ]



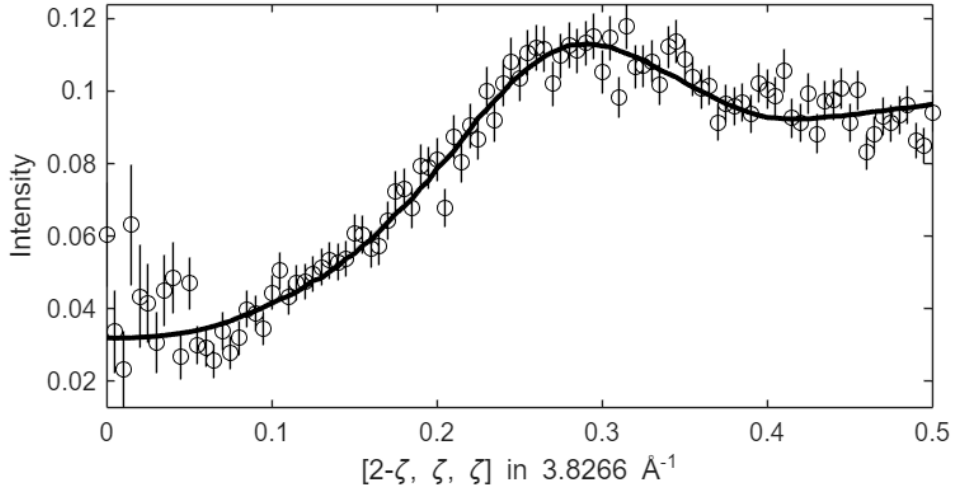
**** En = 243±5 going to 283

Fe_ei800, w2_800MinFF.sqw

S=0.71±0.023; J0= 40±0.19; J1= 0± 0; gamma=74.1991±1.34614

.1 ≤ ξ ≤ 0.1 in [2-ξ, -ξ, 0] , -0.1 ≤ η ≤ 0.1 in [2+0.5η, -0.5η, η] , 238 ≤ E ≤

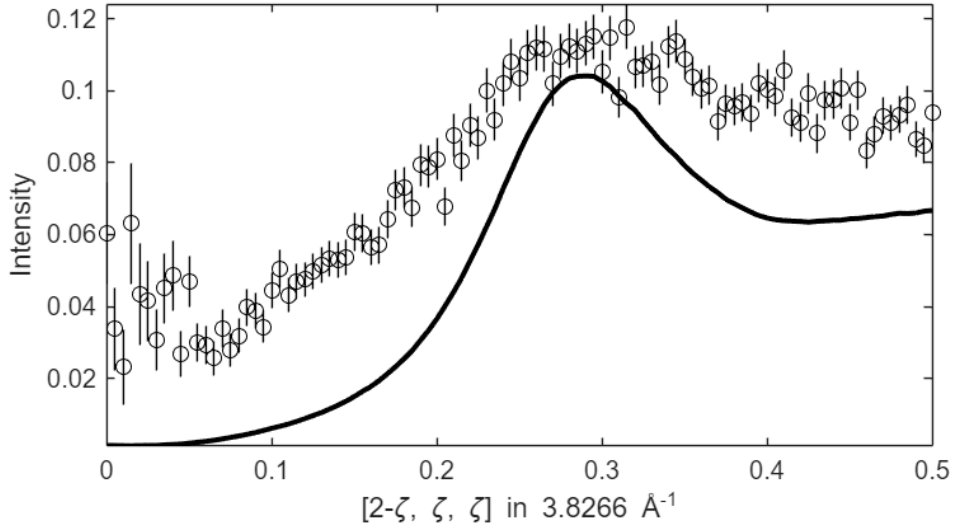
ζ=-0.0025:0.005:0.5025 in [2-ζ, ζ, ζ]



Fe_ei800, w2_800MinFF.sqw

.1 ≤ ξ ≤ 0.1 in [2-ξ, -ξ, 0] , -0.1 ≤ η ≤ 0.1 in [2+0.5η, -0.5η, η] , 238 ≤ E ≤

ζ=-0.0025:0.005:0.5025 in [2-ζ, ζ, ζ]



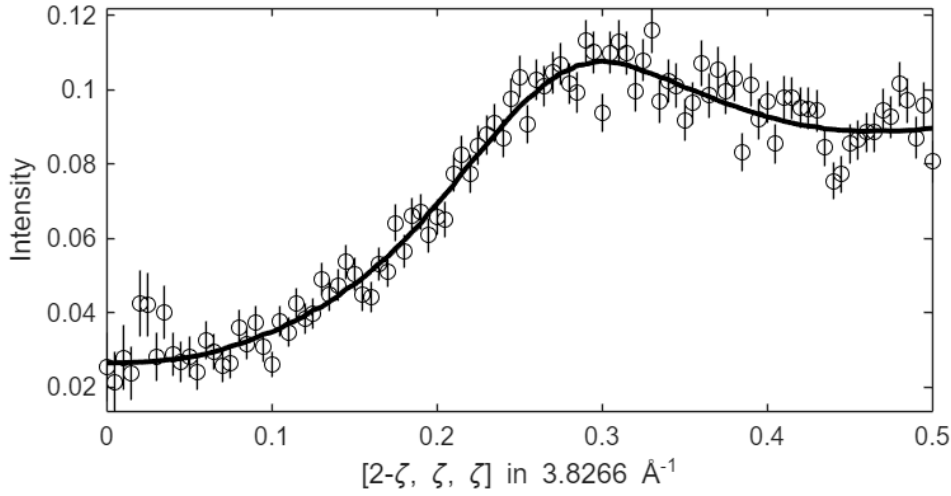
**** En = 253±5 going to 283

Fe_ei800, w2_800MinFF.sqw

S=0.69±0.048; J0= 41±0.46; J1= 0± 0; gamma=74.1285±6.50816

.1 ≤ ξ ≤ 0.1 in [2-ξ, -ξ, 0] , -0.1 ≤ η ≤ 0.1 in [2+0.5η, -0.5η, η] , 248 ≤ E ≤

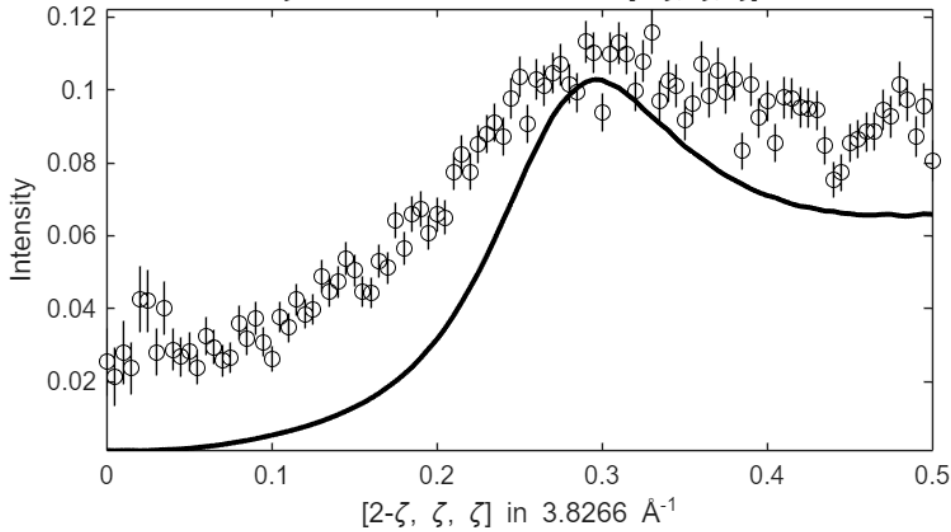
ζ=-0.0025:0.005:0.5025 in [2-ζ, ζ, ζ]



Fe_ei800, w2_800MinFF.sqw

.1 ≤ ξ ≤ 0.1 in [2-ξ, -ξ, 0] , -0.1 ≤ η ≤ 0.1 in [2+0.5η, -0.5η, η] , 248 ≤ E ≤

ζ=-0.0025:0.005:0.5025 in [2-ζ, ζ, ζ]



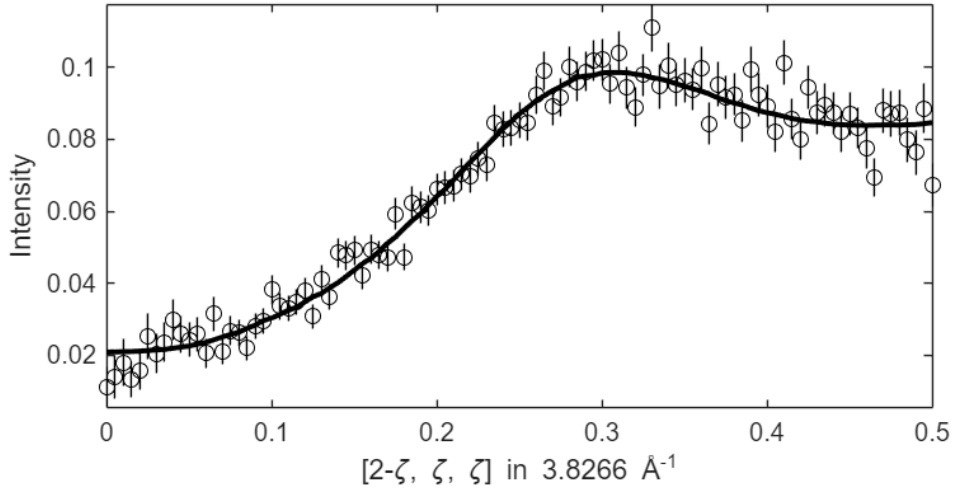
 **** En = 263±5 going to 283

Fe_ei800, w2_800MinFF.sqw

S= 0.8±0.028; J0= 42±0.27; J1= 0± 0; gamma=94.7416±2.38633

.1 ≤ ξ ≤ 0.1 in [2-ξ, -ξ, 0] , -0.1 ≤ η ≤ 0.1 in [2+0.5η, -0.5η, η] , 258 ≤ E ≤

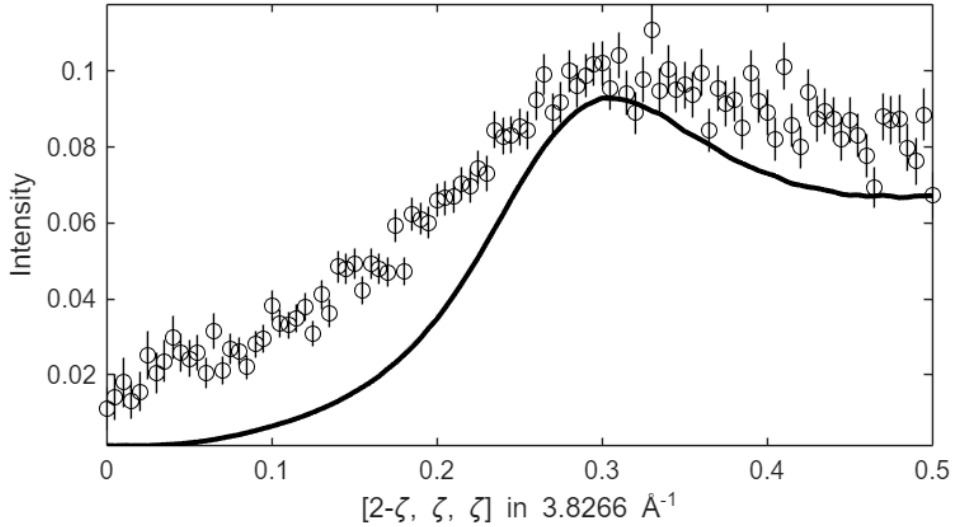
ζ=-0.0025:0.005:0.5025 in [2-ζ, ζ, ζ]



Fe_ei800, w2_800MinFF.sqw

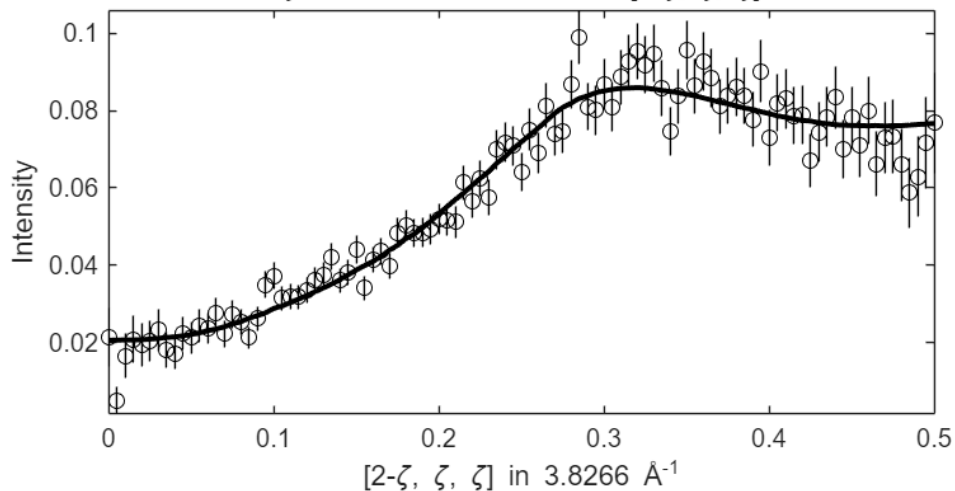
.1 ≤ ξ ≤ 0.1 in [2-ξ, -ξ, 0] , -0.1 ≤ η ≤ 0.1 in [2+0.5η, -0.5η, η] , 258 ≤ E ≤

ζ=-0.0025:0.005:0.5025 in [2-ζ, ζ, ζ]

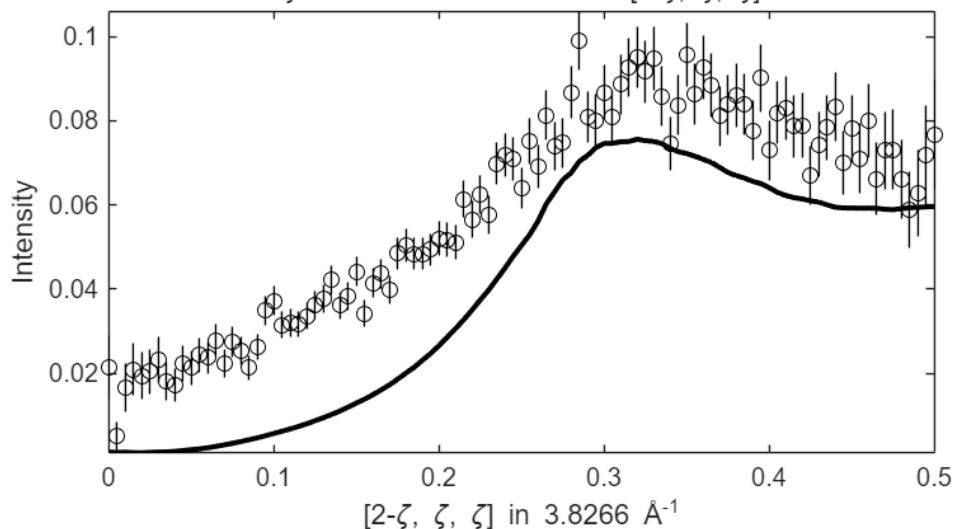


**** En = 273±5 going to 283

Fe_ei800, w2_800MinFF.sqw
 $S=0.75\pm0.068$; $J0=43\pm0.58$; $J1=0\pm0$; $\gamma=106.593\pm12.0538$
 $.1 \leq \xi \leq 0.1$ in $[2-\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2+0.5\eta, -0.5\eta, \eta]$, $268 \leq E \leq$
 $\zeta=-0.0025:0.005:0.5025$ in $[2-\zeta, \zeta, \zeta]$



Fe_ei800, w2_800MinFF.sqw
 $.1 \leq \xi \leq 0.1$ in $[2-\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2+0.5\eta, -0.5\eta, \eta]$, $268 \leq E \leq$
 $\zeta=-0.0025:0.005:0.5025$ in $[2-\zeta, \zeta, \zeta]$



**** En = 283±5 going to 283

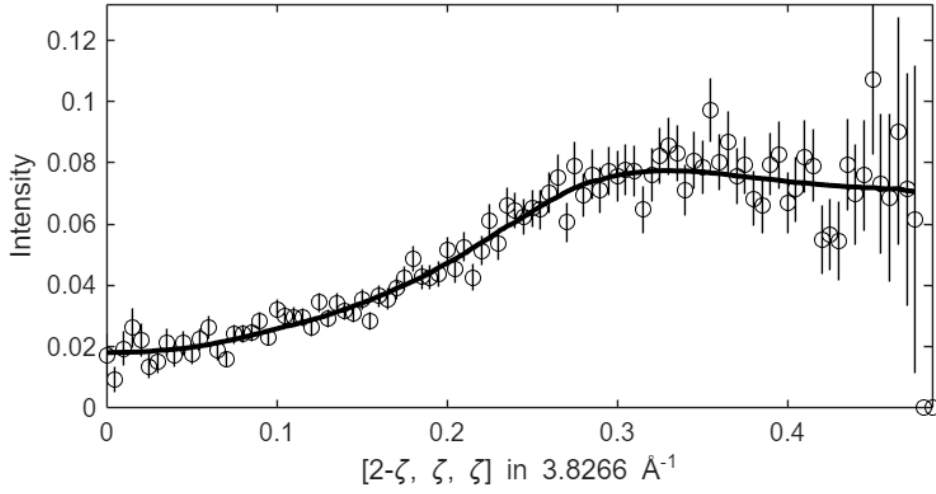
Warning: Dataset:2 points with zero, negative or undefined (infinite or NaN) error bars have been removed from fit, which is 1.9802

Fe_ei800, w2_800MinFF.sqw

S=0.84±0.098; J0= 43±0.76; J1= 0± 0; gamma=131.922±18.3562

.1 ≤ ξ ≤ 0.1 in [2-ξ, -ξ, 0] , -0.1 ≤ η ≤ 0.1 in [2+0.5η, -0.5η, η] , 278 ≤ E ≤

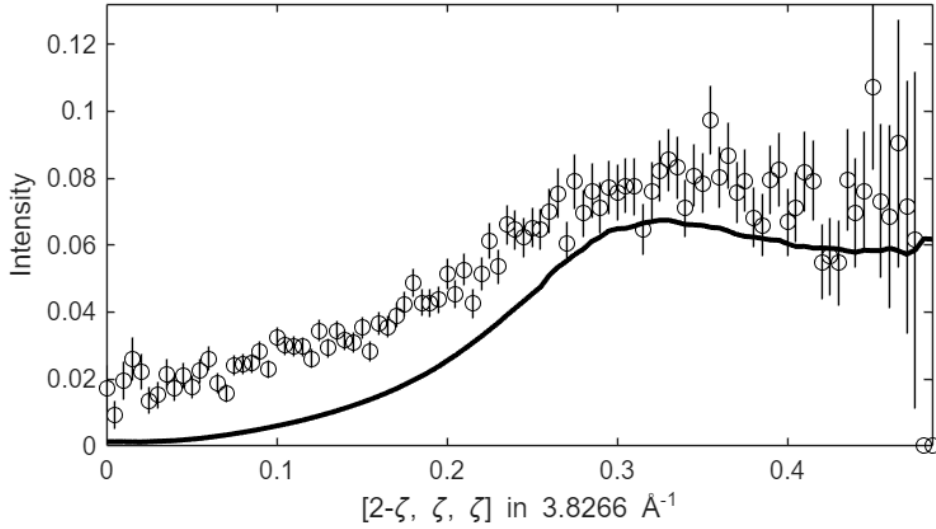
ζ=-0.0025:0.005:0.5025 in [2-ζ, ζ, ζ]

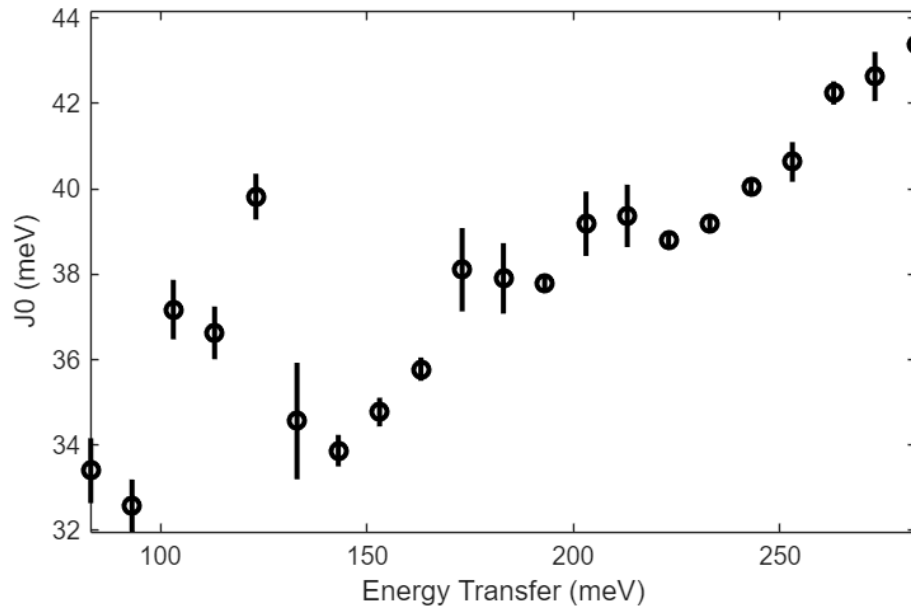
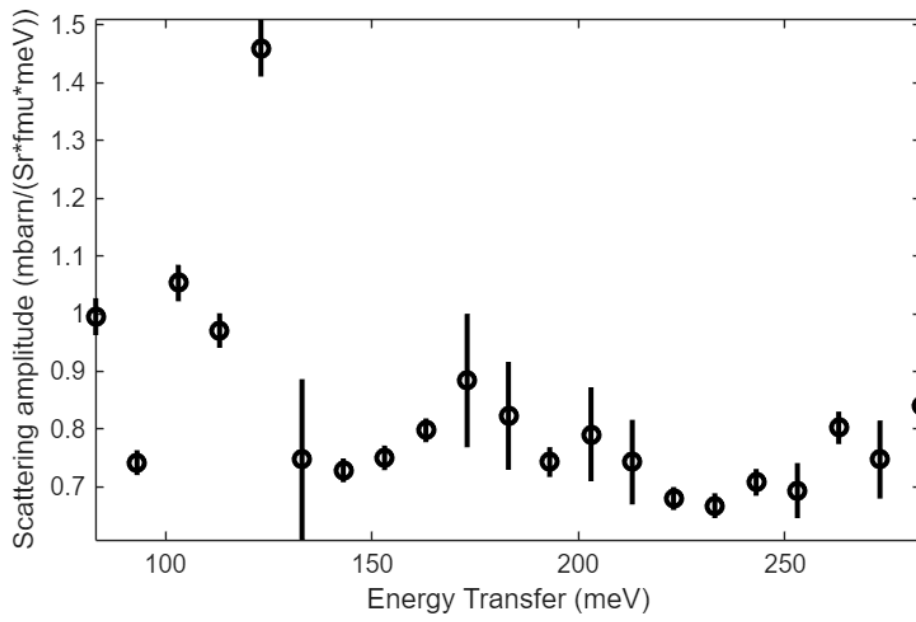


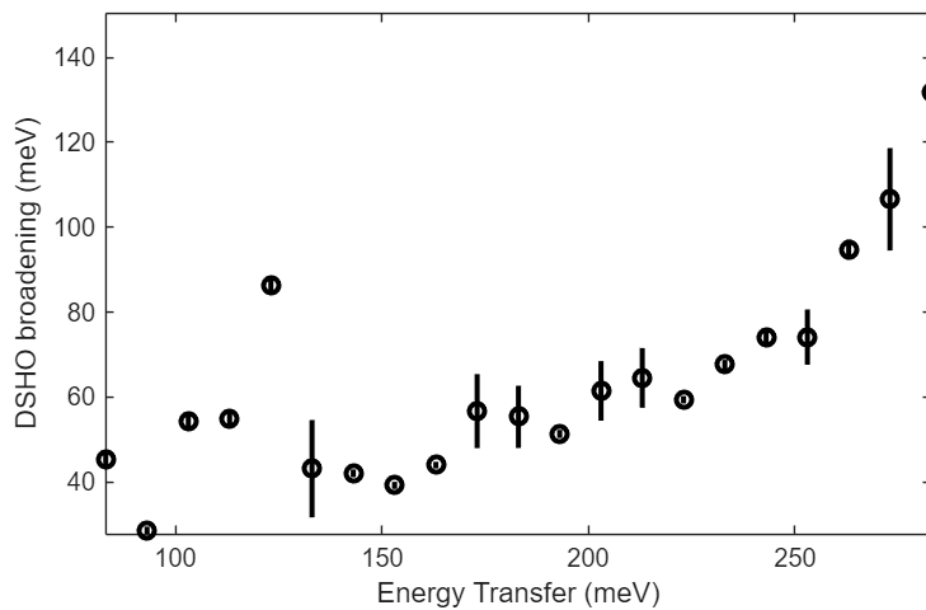
Fe_ei800, w2_800MinFF.sqw

.1 ≤ ξ ≤ 0.1 in [2-ξ, -ξ, 0] , -0.1 ≤ η ≤ 0.1 in [2+0.5η, -0.5η, η] , 278 ≤ E ≤

ζ=-0.0025:0.005:0.5025 in [2-ζ, ζ, ζ]







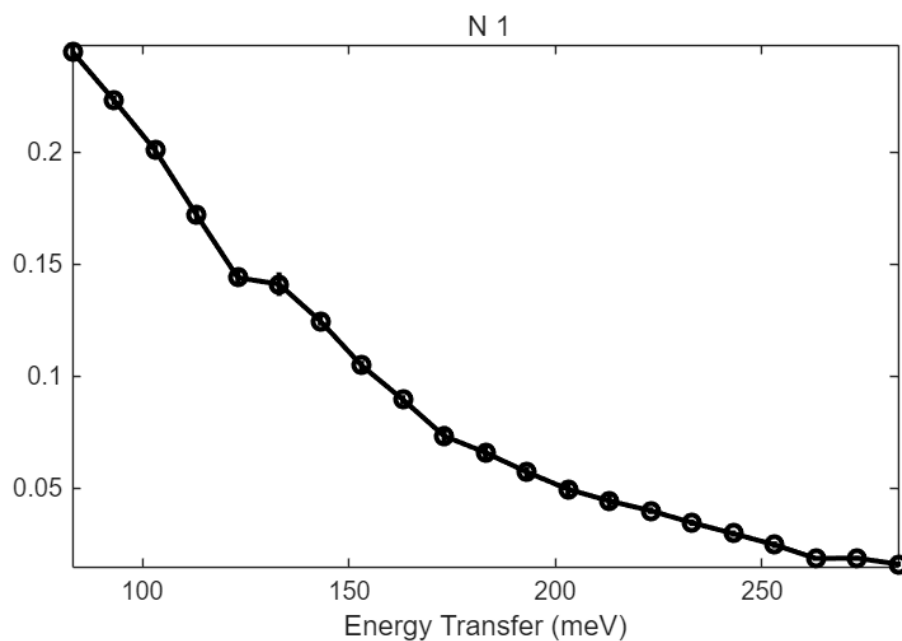
```

c1 = GP_FFCutsFits.(selectCuts);
GP_FFCutsFits.(selectCuts) = [c1{:}];

beep
beep
beep

all_fit_par = GP_FFCutsFits.([selectCuts, '_all_fit_par']);
bg_data = extract_bg_par(all_fit_par,[1,2]);
% original slope parameters
pd(bg_data(1));lx(all_fit_par(1).en,all_fit_par(end).en);keep_figure;

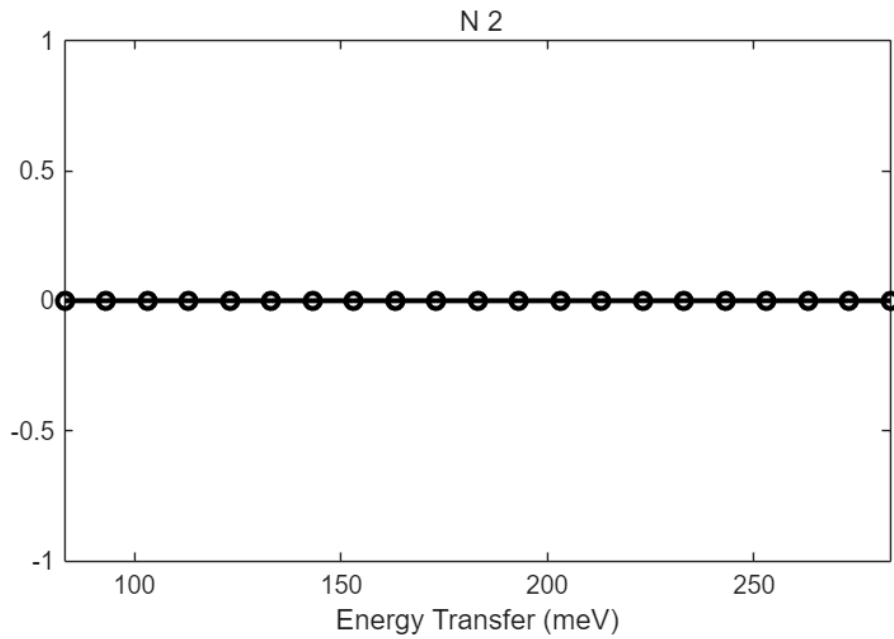
```



```

pd(bg_data(2));lx(all_fit_par(1).en,all_fit_par(end).en);keep_figure

```



```
%pd(bg_data(3));lx(fit_res.peak_fit_par(1).en,fit_res.peak_fit_par(end).en);keep_fig
ure;
%pd(bg_data(4));lx(fit_res.peak_fit_par(1).en,fit_res.peak_fit_par(end).en);keep_fig
ure
```

```
doc = matlab.desktop.editor.getActive;
doc.save;
thisFile = 'e:\SHARE\Fe\Data\analysis\06_fit_with_AQ0\fit_cutFe_GP.mlx';
targ_file = fullfile(fileparts(thisFile),['fit_cutFe_GP_',selectCuts,'.pdf']);
export(thisFile,targ_file);
%close all;

fit_src_struct = GP_FFCutsFits;
save(CutFitFileGP,"fit_src_struct",-v7.3')
%end
```